

Problem Set 1 – Temporal Profile Analyzer

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Source Code

```
1  """
2  Program: Temporal Profile Analyzer
3  Purpose: Computes an AI Era Readiness Score from user metadata (full name
4           and current age), using string processing, explicit type
5           conversion, and the datetime module for the current year.
6  Author: Kshitiz
7  """
8
9  import datetime
10
11  SINGULARITY_YEAR = 2045
12
13  # --- Name Analysis ---
14  user_full_name = input("Enter your full name: ").strip()
15
16  if not user_full_name:
17      print("Input Error: Name cannot be empty or whitespace-only.")
18  else:
19      name_length = len(user_full_name)
20      formatted_name = user_full_name.title()
21      print(f"Identifier Byte-Count: {name_length}")
22      print(f"Formatted Name: {formatted_name}")
23
24  # --- Temporal Projection ---
25  age_input = input("Enter your current age: ").strip()
26
27  if not age_input.isdigit():
28      print("Input Error: Age must be a valid non-negative whole number.")
29  else:
30      current_age = int(age_input)
31      current_year = datetime.date.today().year
32      age_in_2045 = current_age + (SINGULARITY_YEAR - current_year)
33      print(f"Current Year: {current_year}")
34      print(f"Projected Age in {SINGULARITY_YEAR}: {age_in_2045}")
35
36  # --- AI Readiness Formula ---
37  score = ((name_length * 10) + age_in_2045) / 2
38  print(f"AI Era Readiness Score: {score:.2f}")
39
40  # --- Challenge Task (Bonus): Name Repetition ---
41  # For a two-digit age, age // 10 isolates the leading (tens) digit.
42  # For a single-digit age there is no tens digit, so age // 10 would
43  # silently evaluate to 0 and produce an empty string. In that case
44  # we treat the age itself as the repeat count instead.
45  if current_age >= 10:
46      repeat_count = current_age // 10
47  else:
48      repeat_count = current_age
49
50  repeated_name = formatted_name * repeat_count
51  print(f"Bonus - Repeated Name ({repeat_count}x): {repeated_name}")
```

Console Log

```
1 --- Run 1: Perfect run (matches worked example) ---
2 Enter your full name: Scooby-Dooo
3 Identifier Byte-Count: 12
4 Formatted Name: Scooby-Dooo
5 Enter your current age: 20
6 Current Year: 2026
7 Projected Age in 2045: 39
8 AI Era Readiness Score: 79.50
9 Bonus - Repeated Name (2x): Scooby-DoooScooby-Dooo
10
11 --- Run 2: Failed run (non-numeric age) ---
12 Enter your full name: Kshitiz Reddy
13 Identifier Byte-Count: 13
14 Formatted Name: Kshitiz Reddy
15 Enter your current age: twenty
16 Input Error: Age must be a valid non-negative whole number.
```

Submission Notes

The name check runs on the string after `.strip()`, so a name of only spaces is caught by not `user_full_name` before it ever reaches the length calculation. Age is validated with `str.isdigit()` before casting to `int`, so a bad value like "twenty" never reaches `int()` and never raises an error. The current year comes from `datetime.date.today().year` instead of a hard-coded number, so the 2045 projection stays correct no matter when the script runs.

For the bonus, the repeat count is normally the tens digit of the age (`age // 10`). That breaks for single-digit ages since it always gives 0, so the script falls back to using the age itself as the repeat count whenever `current_age < 10`.